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(54) **COMPOSITION FOR REGULATING
PRODUCTION OF INTERFERING
RIBONUCLEIC ACID**

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(57)

ABSTRACT

The embodiments of the present disclosure relate to decreasing the bioavailability of one or more target biomolecules by providing a composition that comprises a recombinant plasmid with one or more sequences of micro interfering ribonucleic acid (miRNA). When the recombinant plasmid interacts with a target cell, it causes the target cell to upregulate production of the miRNA, which then decreases the bioavailability of the target biomolecule. In some embodiments of the present disclosure, the target biomolecule is a kinase.

Specification includes a Sequence Listing.

COMPOSITION FOR REGULATING PRODUCTION OF INTERFERING RIBONUCLEIC ACID

CROSS-REFERENCE TO RELATED APPLICATION

[0001] The present application is a continuation of U.S. patent application Ser. No. 18/518,069 filed Nov. 22, 2023, entitled “COMPOSITION FOR REGULATING PRODUCTION OF INTERFERING RIBONUCLEIC ACID” currently pending, the entirety of which is incorporated herein by reference.

SEQUENCE LISTING

[0002] This application contains a Sequence Listing electronically submitted via Patent Center to the United States Patent and Trademark Office as an XML Document file entitled “A8149358US-Sequence Listing.xml” created on 2023 Nov. 17 and having a size of 75,546 bytes. The information contained in the Sequence Listing is incorporated by reference herein.

TECHNICAL FIELD

[0003] The present disclosure generally relates to compositions for regulating production of interfering ribonucleic acid (RNA). In particular, the present disclosure relates to compositions for regulating gene expression and therefore, the production of interfering RNA, that will suppress over-expression or mis-expression of kinases.

BACKGROUND

[0004] Bioactive molecules, including kinases, are necessary for the homeostatic control of biological systems.

[0005] When bioactive molecules are over-expressed or mis-expressed, homeostasis is lost, and disease is often the result.

[0006] As such, it may be desirable to establish therapies, treatments and/or interventions that address when homeostasis and regulation of bioactive molecules is lost to prevent or treat the resulting disease.

SUMMARY

[0007] Some embodiments of the present disclosure relate to one or more compositions that upregulate the production of one or more sequences of micro interfering ribonucleic acid (miRNA). The sequences of miRNA may be complementary to a sequence of target messenger RNA (mRNA) that encodes for translation of a target biomolecule and the miRNA can cause the bioavailability of the target mRNA to decrease because it is degraded or inactivated by the miRNA, thereby causing a decrease in bioactivity of the target biomolecule within a subject that is administered the one or more compositions. In some embodiments of the present disclosure, the target biomolecule is a kinase. In some embodiments of the present disclosure, the target biomolecule is a kinase such as Bruton's tyrosine kinase. In some embodiments of the present disclosure, the target biomolecule is a kinase such as epidermal growth factor (EGF). In some embodiments of the present disclosure, the target biomolecule is a kinase such as vascular endothelial growth factor (VEGF). In some embodiments of the present disclosure, the target biomolecule is a kinase such as B-Raf.

In some embodiments of the present disclosure, the target biomolecule is a kinase such as anaplastic lymphoma kinase (ALK). In some embodiments of the present disclosure, the target biomolecule is a kinase such as human epidermal growth factor receptor (HER). In some embodiments of the present disclosure, the target biomolecule is a kinase such as Fms-like tyrosine kinase 3 (FLT3). In some embodiments of the present disclosure, the target biomolecule is a kinase such as poly-ADP ribose polymerase (PARP).

[0008] In some embodiments of the present disclosure the compositions comprise a plasmid of deoxyribonucleic acid (DNA) that includes one or more insert sequences of nucleic acids that encode for the production of miRNA and a backbone sequence of nucleic acids that facilitates introduction of the one or more insert sequences into one or more of a subject's cells where it is expressed and/or replicated. Expression of the one or more insert sequences by one or more cells of the subject results in an increased production of the miRNA and, therefore, decreased translation or production of the target biomolecule by one or more of the subject's cells.

[0009] Some embodiments of the present disclosure relate to compositions that upregulate the production of miRNA that degrades, or causes degradation of, or inactivates or causes the inactivation of, the target mRNA of the target biomolecule.

[0010] Some embodiments of the present disclosure relate to a recombinant plasmid (RP). In some embodiments of the present disclosure, the RP comprises a nucleotide sequence of SEQ ID NO. 1 and SEQ ID NO. 2. The RP comprises a nucleotide sequence encoding one or more nucleotide sequences encoding a miRNA sequence that targets mRNA of Bruton's tyrosine kinase.

[0011] Some embodiments of the present disclosure relate to a recombinant plasmid. In some embodiments of the present disclosure, the RP comprises a nucleotide sequence of SEQ ID NO. 1 and SEQ ID NO. 3. The RP comprises a nucleotide sequence encoding one or more nucleotide sequences encoding a miRNA sequence that targets mRNA of EGF.

[0012] Some embodiments of the present disclosure relate to a recombinant plasmid. In some embodiments of the present disclosure, the RP comprises a nucleotide sequence of SEQ ID NO. 1 and SEQ ID NO. 4. The RP comprises a nucleotide sequence encoding one or more nucleotide sequences encoding a miRNA sequence that targets mRNA of VEGF.

[0013] Some embodiments of the present disclosure relate to a recombinant plasmid. In some embodiments of the present disclosure, the RP comprises a nucleotide sequence of SEQ ID NO. 1 and SEQ ID NO. 5. The RP comprises a nucleotide sequence encoding one or more nucleotide sequences encoding a miRNA sequence that targets mRNA of BRAF.

[0014] Some embodiments of the present disclosure relate to a recombinant plasmid. In some embodiments of the present disclosure, the RP comprises a nucleotide sequence of SEQ ID NO. 1 and SEQ ID NO. 6. The RP comprises a nucleotide sequence encoding one or more nucleotide sequences encoding a miRNA sequence that targets mRNA of ALK.

[0015] Some embodiments of the present disclosure relate to a recombinant plasmid. In some embodiments of the present disclosure, the RP comprises a nucleotide sequence

of SEQ ID NO. 1 and SEQ ID NO. 7. The RP comprises a nucleotide sequence encoding one or more nucleotide sequences encoding a miRNA sequence that targets mRNA of HER.

[0016] Some embodiments of the present disclosure relate to a recombinant plasmid. In some embodiments of the present disclosure, the RP comprises a nucleotide sequence of SEQ ID NO. 1 and SEQ ID NO. 8. The RP comprises a nucleotide sequence encoding one or more nucleotide sequences encoding a miRNA sequence that targets mRNA of FLT3.

[0017] Some embodiments of the present disclosure relate to a recombinant plasmid. In some embodiments of the present disclosure, the RP comprises a nucleotide sequence of SEQ ID NO. 1 and SEQ ID NO. 9. The RP comprises a nucleotide sequence encoding one or more nucleotide sequences encoding a miRNA sequence that targets mRNA of PARP.

[0018] Some embodiments of the present disclosure relate to a method of making a composition/target cell complex. The method comprising a step of administering a RP comprising SEQ ID NO. 1 and one of SEQ ID NO. 2, SEQ ID NO. 3, SEQ ID NO. 4, SEQ ID NO. 5, SEQ ID NO. 6, SEQ ID NO. 7, SEQ ID NO. 8, or SEQ ID NO. 9, to a target cell for forming the composition/target cell complex, wherein the composition/target cell complex causes the target cell to increase production of one or more sequences of miRNA that decreases production of a target biomolecule.

[0019] Embodiments of the present disclosure relate to at least one approach for inducing endogenous production of one or more sequences of miRNA that target and silence mRNA of a target biomolecule, for example Bruton's tyrosine kinase. A first approach utilizes gene vectors containing nucleotide sequences for increasing the endogenous production of one or more sequences of miRNA, which are complete or partial sequences and/or combinations thereof, that target and silence the mRNA of Bruton's tyrosine kinase, which can be administered to a subject to increase the subject's production of one or more sequences of the miRNA.

[0020] Embodiments of the present disclosure relate to at least one approach for inducing endogenous production of one or more sequences of miRNA that target and silence mRNA of a target biomolecule, for example EGF. A first approach utilizes gene vectors containing nucleotide sequences for increasing the endogenous production of one or more sequences of miRNA, which are complete or partial sequences and/or combinations thereof, that target and silence the mRNA of EGF, which can be administered to a subject to increase the subject's production of one or more sequences of the miRNA.

[0021] Embodiments of the present disclosure relate to at least one approach for inducing endogenous production of one or more sequences of miRNA that target and silence mRNA of a target biomolecule, for example VEGF. A first approach utilizes gene vectors containing nucleotide sequences for increasing the endogenous production of one or more sequences of miRNA, which are complete or partial sequences and/or combinations thereof, that target and silence the mRNA of VEGF, which can be administered to a subject to increase the subject's production of one or more sequences of the miRNA.

[0022] Embodiments of the present disclosure relate to at least one approach for inducing endogenous production of

one or more sequences of miRNA that target and silence mRNA of a target biomolecule, for example BRAF. A first approach utilizes gene vectors containing nucleotide sequences for increasing the endogenous production of one or more sequences of miRNA, which are complete or partial sequences and/or combinations thereof, that target and silence the mRNA of BRAF, which can be administered to a subject to increase the subject's production of one or more sequences of the miRNA.

[0023] Embodiments of the present disclosure relate to at least one approach for inducing endogenous production of one or more sequences of miRNA that target and silence mRNA of a target biomolecule, for example ALK. A first approach utilizes gene vectors containing nucleotide sequences for increasing the endogenous production of one or more sequences of miRNA, which are complete or partial sequences and/or combinations thereof, that target and silence the mRNA of ALK, which can be administered to a subject to increase the subject's production of one or more sequences of the miRNA.

[0024] Embodiments of the present disclosure relate to at least one approach for inducing endogenous production of one or more sequences of miRNA that target and silence mRNA of a target biomolecule, for example HER. A first approach utilizes gene vectors containing nucleotide sequences for increasing the endogenous production of one or more sequences of miRNA, which are complete or partial sequences and/or combinations thereof, that target and silence the mRNA of HER, which can be administered to a subject to increase the subject's production of one or more sequences of the miRNA.

[0025] Embodiments of the present disclosure relate to at least one approach for inducing endogenous production of one or more sequences of miRNA that target and silence mRNA of a target biomolecule, for example FLT3. A first approach utilizes gene vectors containing nucleotide sequences for increasing the endogenous production of one or more sequences of miRNA, which are complete or partial sequences and/or combinations thereof, that target and silence the mRNA of FLT3, which can be administered to a subject to increase the subject's production of one or more sequences of the miRNA.

[0026] Embodiments of the present disclosure relate to at least one approach for inducing endogenous production of one or more sequences of miRNA that target and silence mRNA of a target biomolecule, for example PARP. A first approach utilizes gene vectors containing nucleotide sequences for increasing the endogenous production of one or more sequences of miRNA, which are complete or partial sequences and/or combinations thereof, that target and silence the mRNA of PARP, which can be administered to a subject to increase the subject's production of one or more sequences of the miRNA.

DETAILED DESCRIPTION

[0027] Unless defined otherwise, all technical and scientific terms used therein have the meanings that would be commonly understood by one of skill in the art in the context of the present description. Although any methods and materials similar or equivalent to those described therein can also be used in the practice or testing of the present disclosure, the preferred methods and materials are now described. All publications mentioned therein are incorporated therein by

reference to disclose and describe the methods and/or materials in connection with which the publications are cited.

[0028] As used therein, the singular forms “a”, “an”, and “the” include plural references unless the context clearly dictates otherwise. For example, reference to “a composition” includes one or more compositions and reference to “a subject” or “the subject” includes one or more subjects.

[0029] As used therein, the terms “about” or “approximately” refer to within about 25%, preferably within about 20%, preferably within about 15%, preferably within about 10%, preferably within about 5% of a given value or range. It is understood that such a variation is always included in any given value provided therein, whether or not it is specifically referred to.

[0030] As used therein, the term “ameliorate” refers to improve and/or to make better and/or to make more satisfactory.

[0031] As used therein, the term “cell” refers to a single cell as well as a plurality of cells or a population of the same cell type or different cell types. Administering a composition to a cell includes in vivo, in vitro and ex vivo administrations and/or combinations thereof.

[0032] As used therein, the term “complex” refers to an association, either direct or indirect, between one or more particles of a composition and one or more target cells. This association results in a change in the metabolism of the target cell. As used therein, the phrase “change in metabolism” refers to an increase or a decrease in the one or more target cells’ production of one or more proteins, and/or any post-translational modifications of one or more proteins.

[0033] As used therein, the term “composition” refers to a substance that, when administered to a subject, causes one or more chemical reactions and/or one or more physical reactions and/or one or more physiological reactions and/or one or more immunological reactions in the subject. In some embodiments of the present disclosure, the composition is a plasmid vector.

[0034] As used therein, the term “endogenous” refers to the production and/or modification of a molecule that originates within a subject.

[0035] As used therein, the term “exogenous” refers to a molecule that is within a subject but that did not originate within the subject. As used therein, the terms “production”, “producing” and “produce” refer to the synthesis and/or replication of DNA, the transcription of one or more sequences of RNA, the translation of one or more amino acid sequences, the post-translational modifications of an amino acid sequence, and/or the production of one or more regulatory molecules that can influence the production and/or functionality of an effector molecule or an effector cell. For clarity, “production” is also used therein to refer to the functionality of a regulatory molecule, unless the context reasonably indicates otherwise.

[0036] As used therein, the term “subject” refers to any therapeutic target that receives the composition. The subject can be a vertebrate, for example, a mammal including a human. The term “subject” does not denote a particular age or sex. The term “subject” also refers to one or more cells of an organism, an in vitro culture of one or more tissue types, an in vitro culture of one or more cell types, ex vivo preparations, and/or a sample of biological materials such as tissue, and/or biological fluids.

[0037] As used therein, the term “target biomolecule” refers to a kinase that is found within a subject. A biomolecule may be endogenous or exogenous to a subject.

[0038] As used therein, the term “target cell” refers to one or more cells and/or cell types that are deleteriously affected, either directly or indirectly, by a dysregulated biomolecule. The term “target cell” also refers to cells that are not deleteriously affected but that are the cells in which it is desired that the composition interacts.

[0039] As used therein, the term “therapeutically effective amount” refers to the amount of the composition used that is of sufficient quantity to ameliorate, treat and/or inhibit one or more of a disease, disorder or a symptom thereof. The “therapeutically effective amount” will vary depending on the composition used, the route of administration of the composition and the severity of the disease, disorder or symptom thereof. The subject’s age, weight and genetic make-up may also influence the amount of the composition that will be a therapeutically effective amount.

[0040] As used therein, the terms “treat”, “treatment” and “treating” refer to obtaining a desired pharmacologic and/or physiologic effect. The effect may be prophylactic in terms of completely or partially preventing an occurrence of a disease, disorder or symptom thereof and/or the effect may be therapeutic in providing a partial or complete amelioration or inhibition of a disease, disorder, or symptom thereof. Additionally, the term “treatment” refers to any treatment of a disease, disorder, or symptom thereof in a subject and includes: (a) preventing the disease from occurring in a subject which may be predisposed to the disease but has not yet been diagnosed as having it; (b) inhibiting the disease, i.e., arresting its development; and (c) ameliorating the disease.

[0041] As used therein, the terms “unit dosage form” and “unit dose” refer to a physically discrete unit that is suitable as a unitary dose for patients. Each unit contains a predetermined quantity of the composition and optionally, one or more suitable pharmaceutically acceptable carriers, one or more excipients, one or more additional active ingredients, or combinations thereof. The amount of composition within each unit is a therapeutically effective amount.

[0042] Where a range of values is provided therein, it is understood that each intervening value, to the tenth of the unit of the lower limit unless the context clearly dictates otherwise, between the upper and lower limit of that range and any other stated or intervening value in that stated range, is encompassed within the disclosure. The upper and lower limits of these smaller ranges may independently be included in the smaller ranges, and are also, encompassed within the disclosure, subject to any specifically excluded limit in the stated range. Where the stated range includes one or both of the limits, ranges excluding either or both of those included limits are also included in the disclosure.

[0043] In some embodiments of the present disclosure, a composition is a recombinant plasmid (RP) for introducing genetic material, such as one or more nucleotide sequences, into a target cell for reproduction or transcription of an insert that comprises one or more nucleotide sequences that are carried within the RP. In some embodiments of the present disclosure, the RP is delivered without a carrier, by a viral vector, by a protein coat, or by a lipid vesicle. In some embodiments of the present disclosure, the vector is an adeno-associated virus vector.

[0044] In some embodiments of the present disclosure, the insert comprises one or more nucleotide sequences that encode for production of at least one sequence of miRNA that decreases the production of target biomolecules. The miRNA may, directly or indirectly, bind to and degrade the target mRNA or otherwise inactivate the target mRNA so that less or none of the target-biomolecule protein is produced.

[0045] In some embodiments of the present disclosure, the target biomolecule is Bruton's tyrosine kinase.

[0046] In some embodiments of the present disclosure, the target biomolecule is EGF.

[0047] In some embodiments of the present disclosure, the target biomolecule is VEGF.

[0048] In some embodiments of the present disclosure, the target biomolecule is BRAF.

[0049] In some embodiments of the present disclosure, the target biomolecule is ALK.

[0050] In some embodiments of the present disclosure, the target biomolecule is HER.

[0051] In some embodiments of the present disclosure, the target biomolecule is FLT3.

[0052] In some embodiments of the present disclosure, the target biomolecule is PARP.

[0053] In some embodiments of the present disclosure, the insert comprises one or more nucleotide sequences that each encode one or more miRNA sequences that may be complementary to and degrade, or cause degradation of, mRNA of the target biomolecule.

[0054] Some embodiments of the present disclosure relate to a composition that can be administered to a subject with a condition that results, directly or indirectly, from the production of a dysregulated biomolecule. When a therapeutically effective amount of the composition is administered to the subject, the subject may change production and/or functionality of one or more biomolecules.

[0055] In some embodiments of the present disclosure, the subject may respond to receiving the therapeutic amount of the composition by changing production and/or functionality of one or more intermediary molecules by changing production of one or more DNA sequences, one or more RNA sequences, and/or one or more proteins that regulate the levels and/or functionality of the one or more intermediary molecules. The one or more intermediary molecules regulate the subject's levels and/or functionality of the one or more biomolecules.

[0056] In some embodiments of the present disclosure, administering a therapeutic amount of the composition to a subject upregulates the production, functionality or both one or more sequences of miRNA that each target the mRNA of one or more target biomolecules. In some embodiments of the present disclosure, there are one, two, three, four, five, or six miRNA sequences that each are complementary to and degrade, or cause degradation of, one biomolecule, such as Bruton's tyrosine kinase, EGF, VEGF, BRAF, ALK, HER, FLT3, or PARP. In some embodiments of the present disclosure, the composition may comprise multiple copies of the same nucleotide sequence of miRNA.

[0057] In some embodiments of the present disclosure, the composition is an RP that may be used for gene therapy. The gene therapy is useful for increasing the subject's endogenous production of one or more sequences of miRNA that target the mRNA of a target biomolecule. For example, the RP can contain one or more nucleotide sequences that cause

increased production of one or more nucleotide sequences that cause an increased production of one or more miRNA sequences that are each complementary to and degrade, or cause degradation of, or inactivate, or cause inactivation of, one biomolecule, such as Bruton's tyrosine kinase, EGF, VEGF, BRAF, ALK, HER, FLT3, or PARP. Increased endogenous expression of the one or more miRNA sequences results in a decreased bioavailability of the desired biomolecule, which may also be referred to as a target biomolecule.

[0058] In some embodiments of the present disclosure, the delivery vehicle of the RP used for gene therapy may be a virus that can be enveloped, or not (unenveloped), replication effective or not (replication ineffective), or combinations thereof. In some embodiments of the present disclosure, the vector is a virus that is not enveloped and not replication effective. In some embodiments of the present disclosure, the vector is a virus of the Parvoviridae family. In some embodiments of the present disclosure, the vector is a virus of the genus Dependoparvovirus. In some embodiments of the present disclosure, the vector is an adeno-associated virus (AAV). In some embodiments of the present disclosure, the vector is a recombinant AAV. In some embodiments of the present disclosure, the vector is a recombinant AAV6.2FF.

[0059] In some embodiments of the present disclosure, the delivery vehicle of the RP used for gene therapy may be a protein coat.

[0060] In some embodiments of the present disclosure, the delivery vehicle of the RP used for gene therapy may be a lipid vesicle.

[0061] The embodiments of the present disclosure also relate to administering a therapeutically effective amount of the composition. In some embodiments of the present disclosure, the therapeutically effective amount of the composition that is administered to a patient is between about 10 and about 1×10^{16} TCID₅₀/kg (50% tissue culture infective dose per kilogram of the patient's body mass). In some embodiments of the present disclosure, the therapeutically effective amount of the composition that is administered to the patient is about 1×10^{13} TCID₅₀/kg. In some embodiments of the present disclosure, the therapeutically effective amount of the composition that is administered to a patient is measured in TPC/kg (total particle count of the composition per kilogram of the patient's body mass). In some embodiments the therapeutically effective amount of the composition is between about 10 and about 1×10^{16} TCP/kg.

[0062] Some embodiments of the present disclosure relate to an adenovirus associated virus (AAV) genome consisting of a RP that when operable inside a target cell will cause the target cell to produce a miRNA sequence that downregulates production of a biomolecule, with examples being Bruton's tyrosine kinase, EGF, VEGF, BRAF, ALK, HER, FLT3, or PARP. The RP is comprised of AAV2 inverted terminal repeats (ITRs), a composite CASI promoter, a human growth hormone (HGH) signal peptide followed by a miRNA expression cassette containing up to six different miRNAs targeting Bruton's tyrosine kinase, EGF, VEGF, BRAF, ALK, HER, FLT3, or PARP, followed by a Woodchuck Hepatitis Virus post-transcriptional regulatory element (WPRE) and an SV40 polyA signal.

(backbone sequence No. 1):

SEQ ID NO. 1

5'

aatcaacctctggattacaaaatttgtgaaagattgactggattcttaactatgttgctccttttacgctatgtggatacgtgctttaatgcctttgt
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3'

SEO ID NO. 2

5'

3'

SEO ID NO. 3

5'

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-continued

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3'

(miRNA expression cassette No. 4 - VEGF):

SEQ ID NO. 4

5'
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SEQ ID NO. 5

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3'

(miRNA expression cassette No. 6 - ALK):

SEQ ID NO. 6

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(miRNA expression cassette No. 8 - FLT3):

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SEQ ID NO: 10 = SEQ ID NO: 1 + SEQ ID NO: 2

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3'

SEQ ID NO: 11 = SEQ ID NO: 1 + SEQ ID NO: 3

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SEQ ID NO: 13 = SEQ ID NO: 1 + SEQ ID NO: 5

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[0063] As will be appreciated by those skilled in the art, because the recombinant plasmid is a circular vector, the one or more sequences of the miRNA expression cassettes may be connected at the 3' end of SEQ ID NO. 1, as shown in SEQ ID NO. 10-17 or at the 5' end of SEQ ID NO. 1.

[0064] As will be appreciated by those skilled in the art, a perfect match of nucleotides with each of the miRNA expression cassette sequences is not necessary in order to have the desired result of decreased bioavailability of the target biomolecule as a result of the target cell producing the miRNA sequence that will bind to and degrade the mRNA of the target biomolecule. In some embodiments of the present disclosure, about 80% to about 100% nucleotide sequence matching with each of the miRNA expression cassettes causes the desired result. In some embodiments of the present disclosure, about 85% to about 100% nucleotide sequence matching with each of the miRNA expression cassettes causes the desired result. In some embodiments of the present disclosure, about 90% to about 100% nucleotide sequence matching with each of the miRNA expression cassettes causes the desired result. In some embodiments of the present disclosure, about 95% to about 100% nucleotide sequence matching with each of the miRNA expression cassettes causes the desired result.

EXAMPLE 1—EXPRESSION CASSETTE

[0065] Expression cassettes for expressing miRNA were synthesized. The synthesized miRNA expression cassettes were cloned into the pAVA-00200 plasmid backbone containing the CASI promoter, multiple cloning site (MCS), Woodchuck Hepatitis Virus post-transcriptional regulatory element (WPRE), and Simian virus 40 (SV40) polyadenylation (polyA) sequence, all flanked by the AAV2 inverted terminal repeats (ITR). pAVA-00200 was cut with the restriction enzymes KpnI and XbaI in the MCS and separated on a 1% agarose gel. The band of interest was excised and purified using a gel extraction kit. Each miRNA expression cassette was amplified by polymerase chain reaction (PCR) using Taq polymerase and the PCR products were gel purified and the bands of interest were also excised and purified using a gel extraction kit. These PCR products contained the miRNA expression cassettes in addition to 15 base pair 5' and 3' overhangs that aligned with the ends of the linearized pAVA-00200 backbone. Using in-fusion cloning, the amplified miRNA expression cassettes are integrated with the pAVA-00200 backbone via homologous recombination. The resulting RP contained the following: 5' ITR, CASI promoter, miRNA expression cassette, WPRE, SV40 polyA and ITR 3'.

SEQUENCE LISTING

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ttactagcac	tcacatggaa	caaatggcct	ctctagaat			6339

The invention claimed is:

1. A composition that comprises a recombinant plasmid (RP) comprising a sequence of nucleotides that is 95-100% identical to SEQ ID NO: 10.

2. The composition of claim 1, wherein the RP is encapsulated in a protein coat, a lipid vesicle, or any combination thereof.

3. The composition of claim 1, wherein the RP is encapsulated in a viral vector.

4. The composition of claim 3, wherein the viral vector is a double stranded DNA virus, a single stranded DNA virus, a single stranded RNA virus, or a double stranded RNA virus.

5. The composition of claim 3, wherein the viral vector is an adeno-associated virus.

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